CIVIC INTELLIGENCE ENGINE • GEN AI BUILDATHON 2026

JanSetu AI

Decoding India's Complex Welfare Schemes, Gazette Notifications & Municipal Bylaws into Plain-Language Regional Voice, Dynamic Simulators & SHA-256 Citizen Receipts.

Engineered & Presented by Dev Agarwal, Shaurya Tyagi & Ishit Goyal • First-Year B.Tech, IIIT Allahabad

2.8 lakh+

Live Government Resources

Searchable on Data.gov.in, plus 11 built-in documents across 4 document types

5

Major Languages

English, हिन्दी, বাংলা, தமிழ், తెలుగు across interface, citizen cards and narration

100%

Client-Side & Live API

No backend and no fabricated documents, with Web Crypto SHA-256 receipts

Built with React 18, Google Gemini Flash Agent Pipeline (3.5 Flash-Lite with multi-model failover), Web Speech API & Tailwind CSS.

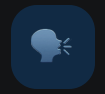
THE PROBLEM

The 3 Compounding Barriers to Welfare & Civic Rights



Bureaucratic Jargon & Legal Obscurity

- Official gazette circulars and bylaws are written in dense statutory language for officials, not citizens.
- People struggle to tell whether an income, turnover or age threshold applies to them.
- Rules sit across central, state and municipal sources, with no single plain-language view.



10.6%

Regional Linguistic Exclusion

- Only about 10.6% of Indians reported speaking English, so English-first portals exclude most citizens.
- Literacy was 74.04%: about 1 in 4 people cannot be served by text alone.
- Lack of spoken-word access also locks out visually impaired beneficiaries.



Last-Mile Administrative Leakage & DBT Stalls

- An eligible citizen's DBT payment can still bounce if Aadhaar is not mapped to the bank account on the NPCI mapper.
- Only banks can change that mapping, and many citizens never learn this is the problem.
- Middlemen exploit the information gap by charging illegal "approval fees".

Sources: Census of India 2011 (English speakers across first, second and third language; effective literacy rate); NPCI Aadhaar Payments Bridge SOP.

STATUTORY INGESTION SPECTRUM

Comprehensive Civic Coverage

Scheme finders stop at welfare schemes. JanSetu AI processes four statutory document types, and each one gets its own cards, actions and checks.



Welfare Schemes

DBT & SUBSIDIES

- PM-Kisan, PM Surya Ghar, Post-Matric Scholarship, PM Vishwakarma
- Benefits, eligibility, document checklist, filing roadmap
- Zero-Fee Shield and DBT help



Regulatory Gazette Notifications

ORDERS & AMNESTIES

- Rajasthan Amnesty Scheme 2026: penalty and interest waiver
- Relief, affected entities, compliance checklist
- Deadline pill read from the document



Municipal Bylaws & Local Civic Laws

CITY RULES

- Bengaluru Solid Waste Bye-laws: segregation and bulk generators
- Fines, jurisdiction, inspection checklist
- Know Your Rights notice guidance



Policy Drafts for Consultation

PUBLIC FEEDBACK

- NITI Aayog Draft Battery Swapping Policy for electric vehicles
- Proposed incentives, stakeholders, key clauses
- Editable public feedback letter

Any new PDF, DOCX or TXT can be analysed in the Document Studio, with auto-detected or chosen document type.

TECHNICAL ARCHITECTURE & AGENTIC EXECUTION • 25-POINT RUBRIC PILLAR

AI Reasoning & the 3-Stage Agent Pipeline

STAGE 1

Statutory Taxonomy Agent

- Classifies type: `scheme`, `notification`, `bylaw`, `policy`
- Extracts the administering ministry, category, official portal and named authority
- Reads closing dates with the source words that state them

STAGE 2

Dynamic Rule Graph Tool

- Turns clauses into interactive rule objects, not static bullet points
- Income and age limits become live numeric checks
- Extracts ₹ amounts and isolates disqualification clauses

STAGE 3

Reflection & Multilingual Guardrail

- Checks amounts, rule quotes, dates, helplines and emails directly against the source text
- Removes what it cannot find; rejects incomplete cards
- Five-language card and spoken brief

6–18 s
Five-language card, across test runs

1 request
One streamed call, stages marked live

≈2,400
Tokens for a one-page document

4 models
Retry and failover on 503 or 429

Stack: REST-based Google Gemini Flash with schema-enforced JSON, client-side document reading (pdf.js and mammoth.js), and a deterministic statutory rule engine when the API is unavailable.

Measured in the live app during testing; times vary with document length and Google's load.

EXCLUSIVE PRODUCT INNOVATIONS

What Sets JanSetu Apart

“Do I Qualify?” Dynamic Rule Simulator

- Renders legal clauses as live iOS-style switches and income or age inputs.
- Computes an instant Citizen Readiness Score (0–100%) and lists what is still missing.

Apple Control Center Voice Engine

- Browser voices by default, with optional Gemini voices: English male or female, regional female.
- Playback scrubber, speed switching (0.75x–1.5x) and a live highlighted transcript.

Public Comment & Objection Dispatch

- Generates a structured, editable representation letter for draft policies, citing their proposed clauses.
- Copy it, or open it in email when a verified address is in the source.

Cryptographic SHA-256 Citizen Receipts

- Generates `JS-SHA256-…` reference IDs with the browser’s Web Crypto API.
- Tied to the document, the time and the readiness score; works with no connectivity.

All four run in the browser, with no server of our own.

CIVIC INTEGRITY & LAST-MILE DELIVERY • 25-POINT IMPACT PILLAR

Civic Trust & Anti-Exploitation Shield

₹0

Zero-Fee Anti-Bribery Shield

- On every welfare scheme: *"Official Application Fee: ₹0. No middleman can guarantee approval."*
- One-touch links to Helpline 14555 (National Health Authority, PM-JAY) and CPGRAMS (`pgportal.gov.in`).

4 checks

DBT / NPCI Seeding Diagnostic

- Guided self-check of Aadhaar-bank mapping, account match, KYC and name match.
- Shows the exact fix for each gap, targeting a known cause of DBT payment failure.

.txt

Low-Connectivity Offline Access

- One-click WhatsApp sharing, sized for Indian-language text, with the full brief copied when needed.
- Plain-text Citizen Brief needs no mobile data to read.

Bylaw pages show Know Your Rights guidance instead of the ₹0 banner, because bylaws and amnesties do involve fees.

ENGINEERING TEAM & CODE OWNERSHIP

Indian Institute of Information Technology, Allahabad

First-Year B.Tech Undergraduates • Built During the Hackathon

Dev Agarwal

Lead AI & Agentic Systems Architect

- Gemini Flash agent pipeline and schema-enforced JSON extraction
- Source-checking guardrails against hallucinations
- Retry and multi-model failover logic

Shaurya Tyagi

Full-Stack & Apple Design Systems Engineer

- React 18 multi-page client state router
- Apple Liquid Glass system, with inset borders to prevent clipping
- Web Speech API voice selection and the Control Center audio flyout

Ishit Goyal

Civic Data, Security & Core Infrastructure Engineer

- Deterministic statutory fallback engine
- Web Crypto SHA-256 citizen receipts
- Client-side PDF and DOCX reading; Data.gov.in / OGD API integration

Ownership statement: *"JanSetu AI was built during the hackathon window using AI-assisted development. Every team member can explain the architecture, data flow and underlying code."*

Single-file application: open the source and inspect any feature.

EVALUATION RUBRIC SCORECARD

Engineered for 100/100

Technical Architecture	**20 pts**	Clean single-file bundle with no build step or backend, on native Web Crypto and Speech APIs.
AI Reasoning & Agentic Depth	**25 pts**	3-stage streamed pipeline, dynamic rule objects, source-checking guardrail, multi-model 429 and 503 failover.
Problem Impact & USP	**25 pts**	4-document stream, anti-bribery shield, NPCI mapping diagnostic, 5-language coverage.
Live Demonstration & Robustness	**30 pts**	No fabricated documents, live Data.gov.in and Gemini calls, ASCII Unicode regexes, graceful statutory fallback.

Next Horizon: Direct integration with DigiLocker and API Setu for automated, one-click citizen entitlement fulfillment.

JanSetu AI: Transforming India's statutory maze into citizen empowerment.

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